

The Linear Regression Model

Columbia University: EC 6411-A

2026

Outline

- 1 Chapter 2: Conditional Expectation
 - Law of Iterated Expectations
 - The Conditional Expectation Function

- 2 Chapter 3: Algebraic Properties of $\hat{\beta}$
 - Orthogonality Conditions and Projections Theorem
 - Goodness of Fit
 - Dummy Variables
 - Multicollinearity

Economic Questions

- does reducing class size improve education outcome?
- is there racial discrimination in the market for home loans?
- does health care spending improve health outcomes?
- by how much will GDP grow in 2024?
- relation between parents' and children's heights.
- effect of job training on earnings

Unconditional Analysis

Goal: provide quantitative answers to some random outcome Y .

- Random variables are summarized by their pdf, f . The mean and variance of f measures the location and dispersion.
 - the unconditional mean of Y is $E[Y] = \int_{-\infty}^{\infty} yf_Y(y)dy \equiv \mu$.
 - the unconditional variance is: $\text{var}(Y) = \int (y - \mu)^2 f_Y(y)dy$.
- Consider a naive model $Y = E[Y] + \overbrace{Y - E[Y]}^e$, $E[e] = 0$.
- But we can do better with information in X .

Conditional Mean and Variance

- With information in X , the conditional mean is

$$\mathbb{E}[Y|X] = \int_{-\infty}^{\infty} y f_{Y|X} dy$$

- the conditional variance is

$$\text{var}(Y|X) = \mathbb{E}\left[\left(Y - \mathbb{E}[Y|X]\right)^2 | X\right].$$

- Decomposition of total variance: If $\mathbb{E}[Y^2] < \infty$,

$$\text{var}(Y) = \mathbb{E}[\text{var}[Y|X]] + \text{var}[\mathbb{E}[Y|X]].$$

Law of Iterated Expectations

- If $\mathbb{E}|Y| < \infty$, for any random $X = (X_1, X_2)$,

$$\mathbb{E}[\mathbb{E}[Y|X_1, X_2]|X_1] = \mathbb{E}[Y|X_1].$$

Note that $\text{var}(Y) \geq \text{var}(Y|X_1) \geq \text{var}(Y|X)$.

- If $\mathbb{E}|Y| < \infty$, then for an random X , $\mathbb{E}[Y] = \mathbb{E}_X[\mathbb{E}[Y|X]]$
- Conditioning Theorem:
 - If $\mathbb{E}|Y| < \infty$, $\mathbb{E}[g(X)Y|X] = g(X)\mathbb{E}[Y|X]$.
 - If, in addition, $\mathbb{E}[g(X)] < \infty$, $\mathbb{E}[g(X)Y] = \mathbb{E}[g(X)\mathbb{E}[Y|X]]$.

The Conditional Expectation Function, $m(X)$

By definition, $Y = \underbrace{E(Y|X)}_{m(X)} + \underbrace{Y - E(Y|X)}_e$.

- If $\mathbb{E}[e|X] = 0$ (mean indep. or conditional mean restriction)
 - For any $h(X)$ such that $\mathbb{E}[h(X)e] < \infty$, $\mathbb{E}[h(X)e] = 0$.
(Implication: e is uncorrelated with any function of X .)
 - Put $h(X) = 1$, $E[e|X] = 0$ implies $\mathbb{E}[e] = 0$
- $\mathbb{E}|e|^r < \infty$ if $\mathbb{E}|Y|^r < \infty$ for $r \geq 1$.
- The conditional expectation of Y given $X = x$ is $E[Y|X = x]$.

- The conditional variance is $\text{var}(e|X) = \sigma^2(X)$.
- If $\mathbb{E}[e^2] < \infty$, the conditional variance of e given $X = x$ is

$$\sigma^2(x) = \text{var}(e|X = x) = \mathbb{E}[(e - \mathbb{E}[e])^2|X = x]$$

- The unconditional variance of e is the average of the conditional variance: $\sigma^2 = \mathbb{E}[e^2] = \mathbb{E}\left[\mathbb{E}[e^2|X]\right] = \mathbb{E}_X[\sigma^2(X)]$

Best Predictor

- Claim: the conditional mean $m(X) = \mathbb{E}[Y|X]$ is the best predictor of Y when given conditioning information X .

Proof: Consider another predictor $g(X)$. Its prediction error is

$$\begin{aligned}\mathbb{E}[(Y - g(X))^2] &= \mathbb{E}[(e + m(X) - g(X))^2] \\ &= \mathbb{E}[e^2] + 2\mathbb{E}[e(m(X) - g(X))] + \mathbb{E}[(m(X) - g(X))^2] \\ &= \mathbb{E}[e^2] + \mathbb{E}[(m(X) - g(X))^2] \\ &\geq \mathbb{E}[e^2] \\ &= \mathbb{E}[(Y - m(X))^2].\end{aligned}$$

But to compute the best predictor $E[Y|X]$, we need to know the density. We consider a linear specification as an approximation.

The Best Linear Predictor

- A linear predictor for Y is a linear function of X . The best predictor minimizes the mean squared prediction error

$$S(b) = \mathbb{E}[(Y - X'b)^2] = \mathbb{E}[Y^2] - 2b'\mathbb{E}[XY] + b'\mathbb{E}[XX']b.$$

- Assumption 2.1: (i) $\mathbb{E}[Y^2] < \infty$; (ii) $\mathbb{E}\|X\|^2 < \infty$ where $\|X\| = (X'X)^{1/2}$; (iii) $\mathbb{E}[XX']$ is positive definite.
- FOC gives $2\mathbb{E}[XY] = 2\mathbb{E}[XX']\beta \Rightarrow \beta = \mathbb{E}[XX']^{-1}\mathbb{E}[XY]$. The linear projection coefficient β is our population parameter.
- The best linear prediction is the linear projection

$$\text{proj}(Y|X) = X'\beta = X'(E[XX'])^{-1}E[XY].$$

The prediction error is $e = Y - X'\beta$ satisfying $\mathbb{E}[Xe] = 0$.

Random Sampling

- In a random sample, n objects are drawn at random from a population with equal probability. Since each object is equally likely to be drawn, the distribution of X_i is the same for all i .
- Denote the i -th random draw by X_i . The random variables $X_1, \dots, X_n$ are iid with marginal distribution F if
 - X_1 is distributed independently of $X_2, \dots, X_n$ and
 - the distribution is the same for all X_i .
- A collection of random vectors $X_1, \dots, X_n$ is a random sample from the population F if X_i are iid.
 - independent: X_i is independent of X_j
 - the vectors X_i, X_j have the same joint distribution

The Linear CEF

- We approximate $m(X) = m(X_1, \dots, X_K)$ by

$$m(X) \approx X_1\beta_1 + X_2\beta_2 + \dots + X_K\beta_K = X'\beta.$$

- This linear CEF is our (parametric) linear regression model

$$\begin{aligned} Y &= X'\beta + e \\ &= X'\beta + \sigma(X)u. \end{aligned}$$

where $u = e/\sigma(X)$ be the rescaled error, $\text{var}(u|X) = 1$.

- e is (conditionally) heteroskedastic if $\sigma^2(x)$ depends on x .
 - otherwise, e is (conditionally) homoskedastic with $\sigma^2(x) = \sigma^2$.
- Goal: estimate the unknown (but non random) β using one realization of the data and then to test hypothesis. We may also be interested in $m(x) = E[Y|X = x]$ as x changes. To do these, we will need to make assumptions.

Definitions (Hansen Probability Chapter 6)

- A parameter θ is any function of the population distribution F .
- A statistic is a function of the sample $x_1, \dots, x_n$.
- An estimator $\hat{\theta}$ for θ is a statistic (rule) for guessing θ .
 - $\hat{\theta}$ is an estimator when we viewed as a function of random obs.
 - It is an estimate when it is calculated from the realized values.

Examples of estimators

- *analog principle approach*:
 - Step 1: express θ as a function of the population F
 - Step 2: express $\hat{\theta}$ as an analog function of the sample
 - Example: $\mu = E[X]$ is the population average value of X . The analog estimator is the sample $\bar{x}_n = \frac{1}{n} \sum_{i=1}^n x_i$.
- *loss function approach*: eg. squared loss, absolute loss.
- *fully parametric*: maximize likelihood.

The Sample Regression Problem

- We have a sample (y_i, x_i) , $i = 1, \dots, n$ from the population. Here, n is the sample size.
 - y_i is the dependent variable or regressand
 - x_i is a $K \times 1$ vector of predictors/covariates/regressors
 - $x_{iK} = 1$.
- Assumption 3.1: $(y_1, x_1), \dots, (y_n, x_n)$ are independent (randomly sampled) and identically distributed (ie. drawn from the same distribution F).
- the *data generating process* is the linear model $y_i = x_i' \beta + e_i$ with $E[e_i | x_i] = E[e_i | x_{i1}, \dots, x_{iK}] = 0$.

Notation

Data in matrix form:

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad X = \begin{pmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1,K-1} & 1 \\ x_{21} & x_{22} & \dots & x_{2,K-1} & 1 \\ \vdots & & & \vdots & \\ x_{n1} & x_{n2} & \dots & x_{n,K-1} & 1 \end{pmatrix}$$

- In Chapter 2: Y is random scalar, X is random vector.
- In Chapter 3: (y_i, x_i) for $i = 1, \dots, n$ is a sample of size n .
- When the joint distribution of (y_i, x_i) does not depend on i , it is the same as that of (Y, X) above, and thus $E[x_i]$ has the same meaning as $E[X]$, and $E[XX']$ is the same as $E[x_i x'_i]$.

Identification

A parameter only can be estimated only if it is identified (uniquely determined).

Example: $Y = X_1\beta_1 + X_2\beta_2 + X_3\beta_3 + X_4\beta_4 + e$.

Suppose $X_4 = X_2 + X_3$:

$$\begin{aligned} Y &= X_1\beta_1 + X_2\beta_2 + X_3\beta_3 + (X_2 + X_3)\beta_4 + e \\ &= X_1\beta_1 + X_2(\beta_2 + \beta_4) + X_3(\beta_3 + \beta_4) + e \end{aligned}$$

We can identify $\beta_2 + \beta_4$ and $\beta_3 + \beta_4$ but not $\beta_2, \beta_3, \beta_4$ separately.

- In the linear model, identification requires X to be linearly independent (full column rank).

Model in Vector and Matrix Form

- Model in scalar form $y_i = x_{i1}\beta_1 + \dots x_{iK}\beta_K + e_i$
- Model for the i th obs.: $y_i = \underbrace{x_i'_{1 \times K}}_{1 \times K} \underbrace{\beta_{K \times 1}}_{K \times 1} + e_i.$
- Model in matrix form: $Y = \underbrace{X_{n \times K}}_{n \times K} \underbrace{\beta_{K \times 1}}_{K \times 1} + e.$
- Population parameter: β unknown but non-random. The goal is estimate β from a sample of data (y_i, x_i) , $i = 1, \dots n$.

Derivation of the Least Squares Estimator

- The OLS estimator minimizes sum of squared residuals

$$S(\beta) = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - x_i' \beta)^2.$$

- the first order conditions are

$$\frac{\partial S(\beta)}{\partial \beta} = -2 \sum_{i=1}^n x_i (y_i - x_i' \beta) = -2X'(y - X\beta).$$

- Let $\hat{\beta}$ satisfy the F.O.C., ie.

$$X'(y - X\hat{\beta}) = 0 \Rightarrow X'y = X'X\hat{\beta}$$

- If $\text{rank}(X)=K$, the inverse of $X'X$ exists. Thus,

$$\hat{\beta} = (X'X)^{-1}X'y = \left(\sum_{i=1}^n x_i x_i' \right)^{-1} \sum_{i=1}^n x_i y_i.$$

Orthogonality condition

- Population errors $e = Y - X\beta$ satisfies $\mathbb{E}[e_i|x_i] = 0$. This implies population orthogonality condition: $E[e_i x_i] = 0$
- Sample residuals: $\hat{e} = Y - X\hat{\beta}$. Recall FOC satisfies

$$\begin{aligned} -2X'Y + 2X'X\hat{\beta} &= 0 \\ X'Y - X'X\hat{\beta} &= 0 \\ \underbrace{X'(Y - X\hat{\beta})}_{\hat{e}} &= \sum_{i=1}^n x_i \hat{e}_i = 0 \end{aligned}$$

- Sample orthogonality condition: $\frac{1}{n} \sum_{i=1}^n \hat{e}_i x_i = 0$, or $\hat{e}'X = 0$
 - system of K equations: $\frac{1}{n} \sum_{i=1}^n \hat{e}_i x_{ik} = 0$ for every k .

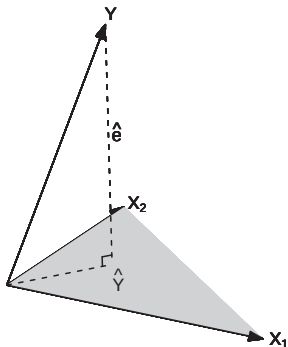
Case $K = 1$, $Y = \beta + e$

- OLS: $\hat{\beta} = (X'X)^{-1}X'Y$ where $X'X = \sum_{i=1}^n x_i x_i'$.
 - Let $\mathbf{1}$ be a $n \times 1$ column of ones.
 - $X'X = \mathbf{1}'\mathbf{1} = \sum_{i=1}^n 1 = n$
 - $X'Y = \mathbf{1}'Y = \sum_{i=1}^n y_i$.
 - $\hat{\beta} = \frac{1}{n} \sum_i y_i = \bar{Y}$.
 - OLS estimates the population mean β by the sample mean.
- Method of Moments
 - $\beta = \mathbb{E}[y_i]$ and $\sigma^2 = \mathbb{E}[y_i^2] - (\mathbb{E}[y_i])^2$.
 - Given a draw of data of size n , we can estimate them by

$$\hat{\beta} = \frac{1}{n} \sum_{i=1}^n y_i, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n y_i^2 - \left(\frac{1}{n} \sum_{i=1}^n y_i \right)^2.$$

Sample moments estimate population moments.

Y , X_1 , and X_2 , which are each elements of $\mathbb{R}^3$. The plane created by X_1 and X_2 is the range space $\mathcal{R}(X)$. Regression fitted values are linear combinations of X_1 and X_2 and so lie on this plane. The fitted value $\hat{Y}$ is the vector on this plane closest to Y . The residual $\hat{e} = Y - \hat{Y}$ is the difference between the two. The angle between the vectors $\hat{Y}$ and $\hat{e}$ is 90° , and therefore they are orthogonal as shown.



Projection Matrices

$$Y = X\beta + e$$

$$\begin{aligned}\hat{e} &= Y - X\hat{\beta} = Y - X(X'X)^{-1}X'Y \\ &= (I_n - X(X'X)^{-1}X')Y = (I_n - P)Y \\ &= MY = Me\end{aligned}$$

- Properties of *idempotent* projection matrices
 - $P = PP$ and $M = MM$.
 - symmetric: $P = P'$, $M = M'$.
 - eigenvalues of P and M are 1 or 0.
 - $\text{trace}(P) = K$, $\text{rank}(P) = K$
 - $\text{trace}(M) = n - K$, $\text{rank}(M) = n - K$.
- P projects onto space spanned by X , hat matrix, $\hat{Y} = PY$
- M projects onto space orthogonal to X : residual maker matrix, $\hat{e} = MY$.

Projections theorem: there exists $\hat{Y}$ and $\hat{e}$ such that

$$Y = \hat{Y} + \hat{e}, \quad \hat{Y}'\hat{e} = 0.$$

- $\hat{Y} = X\hat{\beta}$ where $\hat{\beta} = (X'X)^{-1}X'Y$
- X is orthogonal to $\hat{e}$ and therefore $\hat{Y}$ is orthogonal to $\hat{e}$.
- $\hat{e}$ is mean zero when $x_{ik} = 1$ for all i .
- mean of $\hat{Y}$ equals mean of Y .

Frish-Waugh-Lovell theorem:

- Regress y on X_1 on X_2 to get $\hat{\beta}_1, \hat{\beta}_2$
 - β_2 is the marginal effect of x_2 on Y holding x_1 fixed.
- Consider an estimator $\tilde{\beta}_2$ that partials out the effects of X_1 from Y and X_2 .
 - Regress y on X_1 to get residuals $\tilde{y}$
 - Regress X_2 on X_1 to get residuals $\tilde{X}_2$
 - Regress $\tilde{y} = M_1 Y$ on $\tilde{X}_2 = M_1 X_2$ to get $\tilde{\beta}_2$.
- FWL theorem: $\hat{\beta}_2 = \tilde{\beta}_2$.

Idea: $M_1 Y = M_1 X_1 \beta_1 + M_1 X_2 \beta_2 + M_1 e$ and $M_1 X_1 = 0$.

Implications of FWL:

- 1 $\hat{\beta}_2$ is the same estimate obtained when y is regressed on X_2 alone only when X_1 and X_2 are orthogonal (ie. $X_1'X_2 = 0$).
 - $\tilde{X}_2'\tilde{Y} = X_2'M_1M_1Y = \tilde{X}_2'Y$ since M_1 is idempotent.
 - $\tilde{X}_2 = M_1X_2 = X_2 - X_1(X_1'X_1)^{-1}X_1'X_2 = X_2$ since $X_1'X_2 = 0$
 - $(\tilde{X}_2'\tilde{X}_2)^{-1}\tilde{X}_2'Y = (X_2'X_2)^{-1}X_2'Y$ when $X_1'X_2 = 0$.

2 The parameters β_1, β_2 in the model $Y = \beta_1 + X\beta_2 + e$ can be estimated by either including an intercept, or performing a regression in demeaned data.

- Now $M_1 = I_n - 1(1'1)^{-1}1'$ where 1 is a unit (column) vector being the first column of X . Then $X^* = M_1X = X - \bar{X}$ is a matrix of deviations of X from the sample mean.
- From $\bar{Y} = \hat{\beta}_1 + \bar{X}\hat{\beta}_2$, we get $\hat{\beta}_1 = \bar{Y} - \bar{X}\hat{\beta}_2$.

Goodness of Fit

Recall $Y = \hat{Y} + \hat{e}$.

- $Y'Y = \hat{Y}'\hat{Y} + \hat{e}'\hat{e}$, $R^2 = 1 - \frac{\hat{e}'\hat{e}}{Y'Y}$. But this (uncentered) R^2 is not invariant to changes to the mean of Y .
- Consider $M_1 Y = M_1 \hat{Y} + M_1 \hat{e}$. Since $\hat{e}'\hat{Y} = 0$ and $M_1 \hat{e} = \hat{e}$,

$$\begin{aligned} \underbrace{Y'M_1 Y}_{\text{Total SSR}} &= \underbrace{\hat{\beta}'X'M_1 X\hat{\beta}}_{\text{explained}} + \underbrace{\hat{e}'\hat{e}}_{\text{SSE}}. \\ \text{(centered) } R^2 &= \frac{\hat{\beta}'X'M_1 X\hat{\beta}}{Y'M_1 Y} = 1 - \frac{\hat{e}'\hat{e}}{Y'M_1 Y}. \end{aligned}$$

- $0 \leq R^2 \leq 1$ measures the proportion of variation in Y explained by variation in the regressors.

Adjusted R^2

- R^2 may not be between 0 and 1 if the regression has no constant. It can be negative!
- R^2 will never decrease when a regressor is added, This favors big models with more regressors. Let

$$\bar{R}^2 = 1 - \frac{\frac{1}{n-K} \hat{e}' \hat{e}}{\frac{1}{n-1} Y' M_1 Y}.$$

$\bar{R}^2$ will fall when t ratio of deleted variable exceeds 1.

- $\bar{R}^2$ and R^2 of different models can be compared only if the dependent variable is the same.
- these R^2 measures assume estimator is OLS.

Is linearity restrictive?

In a linear model, $\frac{\partial}{\partial x} m(x) = \beta$ does not depend on x .

- $y_i = Ax_i^\beta \exp(e_i)$ implies log-linear model

$$\log y_i = \underbrace{\alpha}_{\log A} + \beta \log x_i + e_i$$

- semi-log model for growth rates: $\frac{\partial \log Y_t}{\partial t} = \beta$

$$\log Y_t = \beta t + e_t$$

- Interaction effect model:

$$m(x_{i1}, x_{i2}) = x_{i1}\beta_1 + x_{i2}\beta_2 + x_{i1}^2\beta_3 + x_{i2}^2\beta_4 + x_{i1}x_{i2}\beta_5 + \beta_6$$

$$\frac{\partial}{\partial x_{i1}} m(x_{i1}, x_{i2}) = \beta_1 + 2x_{i1}\beta_3 + x_{i2}\beta_5$$

$$\frac{\partial}{\partial x_{i2}} m(x_{i1}, x_{i2}) = \beta_2 + 2x_{i2}\beta_4 + x_{i1}\beta_5.$$

Dummy variables

Dummy variable: Let

$$D_{i1} = \begin{cases} 1 & \text{female} \\ 0 & \text{male} \end{cases} \quad D_{i2} = \begin{cases} 0 & \text{female} \\ 1 & \text{male} \end{cases}$$

Interested in $E(w_i | \text{gender})$, we can use the regression model

$$w_i = \beta_0 + \beta_1 D_{i1} + e_i.$$

Method 1:

$$\begin{aligned} E(\text{wage} | \text{male}) &= \beta_0 \\ E(\text{wage} | \text{female}) &= \beta_0 + \beta_1, \end{aligned}$$

and a wage gap of $-\beta_1$.

Method 2:

$$w_i = \beta_0 + \beta_2 D_{i2} + e_i.$$

The model implies

$$\begin{aligned} E(\text{wage}|\text{male}) &= \beta_0 + \beta_2 \\ E(\text{wage}|\text{female}) &= \beta_0 \end{aligned}$$

Wage gap = β_2 , t test.

Method 3:

$$w_i = b_1 D_{i1} + b_2 D_{i2} + e_i.$$

In this model,

$$E(\text{wage}|\text{female}) = b_1$$

$$E(\text{wage}|\text{male}) = b_2$$

Wage gap: $b_2 - b_1$. Wald test.

Three equivalent ways of obtaining the wage gap.

Dummy Variable Trap we cannot include an intercept, D_{i1} and D_{i2} at the same time since $D_{i1} + D_{i2} = 1$.

Seasonal Dummies

Let $D_{jt} = 1$ if t is in quarter j .

$$y_t = \beta_0 + \beta_1 + \beta_2 x_t + \delta_1 D_{1t} + \delta_2 D_{2t} + \delta_3 D_{3t} + e_t$$

- There are four quarters, we can only include three dummies.
- If we include all 4 dummies, we cannot include an intercept because $\sum_j D_{jt} = 1$.

Example Multiple Dummies

If there are p dummies, CEF takes at most 2^p distinct values.

- $x_{i1} = 1$ if woman, $x_{i2} = 1$ if married.
- Let μ_{jk} be value if $x_{i1} = j$ and $x_{i2} = k$.
- Regression: $y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + \beta_4 + e$.

$$\mathbb{E}[y_i | x_{i1}, x_{i2}] = \begin{cases} \mu_{00} & x_{i1} = 0, x_{i2} = 0 \text{ (unmarried man)} \\ \mu_{01} & x_{i1} = 0, x_{i2} = 1 \text{ (married man)} \\ \mu_{10} & x_{i1} = 1, x_{i2} = 0 \text{ (unmarried woman)} \\ \mu_{11} & x_{i1} = 1, x_{i2} = 1 \text{ (married woman)} \end{cases}$$

- $\beta_1 = \mu_{10} - \mu_{00}$: gender effect on wages among unmarried,
- $\beta_2 = \mu_{01} - \mu_{00}$: effect of marriage amongst man,
- $\beta_3 = \mu_{11} - \mu_{10} - \mu_{01}$: interaction effect
- $\beta_4 = \mu_{00}$.

Example Splines

- Suppose we have 3 age groups: $age < 18$, $18 \leq age \leq 22$, $age > 22$ and $\mathbb{E}[income|age] = \alpha_j + \beta_j age$ for $j = 1, 2, 3$
- Let $D_{i1} = 1$ if $age_i \geq t_1^* = 18$ and $D_{i2} = 1$ if $age_i \geq t_2^* = 22$,

- The natural regression is

$$income = \beta_1 + \beta_2 age + \gamma_1 D_1 + \delta_1 D_1 age + \gamma_2 D_2 + \delta_2 D_2 age + e.$$

But the function would be discontinuous at the kink points.

$$\beta_1 + \beta_2 t_1^* = (\beta_1 + \gamma_1) + (\beta_2 + \delta_1) t_1^*$$

$$\Rightarrow \gamma_1 + \delta_1 t_1^* = 0$$

$$(\beta_1 + \gamma_1) + (\beta_2 + \delta_1) t_2^* = (\beta_1 + \gamma_1 + \gamma_2) + (\beta_2 + \delta_1 + \delta_2) t_2^*$$

$$\Rightarrow \gamma_2 = -\delta_2 t_2^*.$$

- Imposing these restrictions, we obtain

Rubin's Potential Outcomes Framework

- Let $D_i = 1$ if treated, $D_i = 0$ if not treated (control).
- We observe outcome $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$ where potential outcomes are
 - $Y_i(1) = h(1, X_i, U_i)$ if treated
 - $Y_i(0) = h(0, X_i, U_i)$ if control (not treated)

X_i are covariates and let U be unobserved and random.

- Causal effect of D_i on Y_i : $\delta(X_i, U_i) = Y_i(1) - Y_i(0)$.
 - The conditional average causal effect is $E[\delta(X_i, U_i) | X_i = x]$
 - The unconditional average causal effect is $E[\delta(X_i, U_i)]$

Regression Setup

With no covariates and assuming homogeneity,

$$\begin{aligned} Y_i &= Y_i(0) + D_i(Y_i(1) - Y_i(0)) \\ &= E[Y_i(0)] + \underbrace{(Y_i(1) - Y_i(0))}_{\beta} D_i + \underbrace{Y_i(0) - E[Y_i(0)]}_{e_i} \\ &= \alpha + \beta D_i + e_i. \end{aligned}$$

If $E[e_i|D_i] = 0$, we can estimate α and β by OLS. But we need to put additional assumptions for this condition to be true.

- $\delta = E[Y_i|D_i = 1] - E[Y_i|D_i = 0]$ becomes

$$\underbrace{E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 1]}_{\text{ATT of treated}} + \underbrace{E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0]}_{\text{selection bias}}$$

- But if D_i independent of Y_i (random assignment), we can swap $E[Y_i(0)|D_i = 1]$ for $E[Y_i(0)|D_i = 0]$ so that

$$\begin{aligned} E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0] &= E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 1] \\ &= E[Y_i(1) - Y_i(0)] \end{aligned}$$

Random assignment removes selection bias.

- LaLonde 1986: Effect of job training on earnings
 - Units are randomly assigned to the job training program
 - Data: earnings in 1978 Y , other pre-treatment covariates X .

Multicollinearity

The variance of $\hat{\beta}|X$ is $\sigma^2(X'X)^{-1}$. Consider bivariate case when X is demeaned:

$$\begin{aligned}
 \frac{1}{n} \sum_i x_i x_i' &= \begin{pmatrix} \widehat{\text{var}}(x_{i1}) & \widehat{\text{cov}}(x_{i1}, x_{i2}) \\ \widehat{\text{cov}}(x_{i1}, x_{i2}), & \widehat{\text{var}}(x_{i2}) \end{pmatrix} \\
 &= \begin{pmatrix} \hat{\sigma}_1 & 0 \\ 0 & \hat{\sigma}_2 \end{pmatrix} \begin{pmatrix} 1 & \hat{\rho}_{12} \\ \hat{\rho}_{12} & 1 \end{pmatrix} \begin{pmatrix} \hat{\sigma}_1 & 0 \\ 0 & \hat{\sigma}_2 \end{pmatrix} \\
 n \cdot \widehat{\text{var}}(\hat{\beta}|X) &= \sigma^2 \left(\frac{X'X}{n} \right)^{-1} \\
 &= s^2 \begin{pmatrix} \hat{\sigma}_1^{-2} & 0 \\ 0 & \hat{\sigma}_2^{-2} \end{pmatrix} \frac{1}{1 - \hat{\rho}_{12}^2} \begin{pmatrix} 1 & -\hat{\rho}_{12} \\ -\hat{\rho}_{12} & 1 \end{pmatrix} \\
 n \cdot \widehat{\text{var}}(\hat{\beta}_1|X) &= s^2 \hat{\sigma}_1^{-2} \frac{1}{1 - \hat{\rho}_{12}^2}
 \end{aligned}$$

- If $|\hat{\rho}_{12}| = 1$, the variance is infinite.
- If $|\rho_{12}|$ is close to 1, near non-invertibility of $\frac{1}{n} \sum_{i=1}^n x_i x_i'$ makes the variance of $\hat{\beta}_1$ unstable.
- conditioning number: square root of ratio of largest to smallest eigenvalue of $X'X$ (after standardizing X)
- values in excess of 20 suggests a problem.
- remedy: drop some of the highly correlated variables